# Project Salsabil

*Sovereign Satellite Intelligence for*

*Precision Irrigation & Climate Resilience*


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# Abstract


Agriculture in Tunisia accounts for nearly 80% of the nation's water consumption, yet it remains highly vulnerable to climate-induced water scarcity. Project Salsabil addresses this challenge by developing a sovereign, satellite-driven precision irrigation system tailored to Tunisia's unique agricultural landscape. This report outlines a three-phase transition from ground-based calibration to a fully autonomous space-based intelligence infrastructure.

The methodology utilizes a data fusion framework that integrates thermal satellite imagery, vegetation indices, and localized weather data to compute the Crop Water Stress Index (CWSI). Phase 1 establishes a national "digital twin" using multi-source Earth observation (Sentinel-2, Landsat-9, MODIS) and a ground-truth network of IoT sensors across key regions like Sfax and Cap Bon. Phase 2 deploys a physics-informed machine learning engine to deliver real-time irrigation recommendations via SMS and digital dashboards. The final phase involves the launch of a sovereign 6U CubeSat constellation, ensuring Tunisia's strategic autonomy and high-frequency monitoring capabilities.

The expected impact includes a 15–30% reduction in agricultural water waste and a 5–12% increase in yield stability for high-value crops. By establishing a national thermal stress reference library, Project Salsabil strengthens Tunisia's climate resilience and positions the country as a regional provider of agri-space intelligence.

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# Chapter 1

# Introduction

Agriculture in Tunisia lies at the core of both economic stability and food security, yet it operates under increasing environmental constraints. As a semi-arid country, Tunisia faces structural water scarcity, a challenge that has intensified in recent years due to climate change. Rising temperatures, prolonged drought periods, and irregular rainfall patterns are disrupting traditional agricultural cycles and placing significant stress on already limited water resources. In this context, improving the efficiency of water use in agriculture is no longer optional—it is a national priority.

With agriculture consuming nearly 80% of Tunisia's available water resources, the sector represents the largest leverage point for addressing the country's water deficit. However, current irrigation practices remain largely reactive and uniform, relying on fixed quotas or farmer intuition rather than real-time crop needs. This results in substantial inefficiencies, including over-irrigation, energy waste in water pumping, and reduced crop productivity. To ensure long-term sustainability, Tunisia must transition toward a more intelligent, data-driven model of water management.

Recent advances in satellite-based Earth observation provide a unique opportunity to address this challenge. By enabling continuous monitoring of land surface temperature, vegetation health, and environmental conditions, satellite data can reveal early signs of crop stress that are invisible to the human eye. When combined with ground-based measurements and advanced modeling techniques, this data can form the foundation of a predictive system capable of guiding irrigation decisions with high precision.

## 1.1 The Context

Tunisia's agricultural landscape is highly diverse, ranging from olive groves in Sfax to irrigated horticulture in Cap Bon and cereal production in the central regions such as Kairouan and Siliana. Each of these systems exhibits distinct water requirements and responses to environmental stress. Despite this variability, irrigation practices are often applied uniformly, without accounting for crop-specific or region-specific conditions. This mismatch between water allocation and actual plant demand contributes to both resource depletion and suboptimal yields.

The country's increasing exposure to climate variability further complicates water management. Reduced precipitation and higher evapotranspiration rates are accelerating the depletion of groundwater reserves and reducing the reliability of surface water sources. As a result, farmers are forced to operate under growing uncertainty, often reacting to stress only after visible symptoms such as wilting or reduced growth appear. By that stage, the impact on yield is already significant.

Existing monitoring systems are limited in both spatial resolution and responsiveness. Field inspections are labor-intensive and cannot scale to national coverage, while meteorological data alone does not provide direct insight into plant-level water stress. Although satellite data is increasingly available, its operational use in Tunisia remains constrained by a lack of calibration, integration, and actionable delivery mechanisms.

To overcome these limitations, there is a need for a system that combines satellite observations with ground-truth data and advanced modeling to provide reliable, real-time insights into crop water status. Such a system would enable a shift from reactive irrigation practices to proactive, precision-based water management.

## 1.2 The Objective

The primary objective of Project Salsabil is to develop a sovereign, satellite-driven precision irrigation system tailored specifically to Tunisia's agricultural and climatic conditions. The system aims to detect crop water stress in real time and deliver actionable insights that optimize irrigation practices across the country.

This objective is supported by the following specific goals:

1. **Technical Objective:** Develop a data fusion framework integrating thermal satellite imagery, vegetation indices, and localized weather and soil data to accurately compute the Crop Water Stress Index (CWSI) for Tunisian crops.

2. **Calibration Objective:** Build a national "ground truth" network, including IoT-based soil moisture sensors and crop-specific thermal profiles, to ensure that satellite-derived measurements reflect real field conditions in regions such as Sfax, Cap Bon, Kairouan, and Siliana.

3. **Operational Objective:** Provide real-time irrigation recommendations to farmers and regional authorities (CRDAs) through accessible platforms, including SMS alerts and decision-support dashboards.

4. **Sovereignty Objective:** Establish Tunisia's independence in agricultural monitoring by transitioning from externally sourced satellite data to a nationally owned Low Earth Orbit (LEO) satellite constellation.

5. **Economic Objective:** Reduce irrigation water waste, lower energy consumption associated with pumping, and increase agricultural productivity, generating measurable economic benefits at both farm and national levels.

## 1.3 The Value Proposition

Project Salsabil delivers a Tunisia-specific solution to water scarcity by transforming how irrigation decisions are made. Its core value lies in shifting from generalized, reactive practices to precise, data-driven interventions tailored to local crop and environmental conditions.

A central differentiator is **data sovereignty**. By ultimately owning its satellite infrastructure, Tunisia eliminates dependence on foreign data providers and gains full control over the acquisition, processing, and application of agricultural intelligence. This ensures long-term reliability and strategic autonomy in managing critical resources.

The system also introduces **predictive, AI-driven irrigation management**. By combining thermal satellite data with physical modeling, it detects early signs of plant water stress—referred to as "thermal anomalies"—before they become visible. This enables farmers to act proactively, improving both water efficiency and crop health.

Another key advantage is its **adaptation to local agricultural diversity**. Through the development of crop-specific thermal profiles, the system accounts for the unique behavior of different crops and regions within Tunisia. This level of specificity significantly enhances the accuracy and relevance of the recommendations provided.

Finally, Project Salsabil supports **national climate resilience**. By optimizing water use and improving the adaptability of agricultural systems, it strengthens Tunisia's ability to cope with increasing environmental variability. In the long term, the project also creates opportunities for regional expansion, positioning Tunisia as a provider of agri-space intelligence services to neighboring countries facing similar challenges.

# Chapter 2

# Overview of the Solution

Project Salsabil is built as a layered decision-support system that translates satellite observations into irrigation intelligence for Tunisia. Its architecture begins with physical measurements of crop and soil conditions, then moves through calibration and modeling, and finally ends with actionable delivery to farmers and agricultural authorities. The solution is designed to work at multiple scales: from individual fields in regions such as Sfax, Cap Bon, Kairouan, and Siliana, to regional planning by CRDAs, and eventually to national sovereignty through a Tunisian-owned satellite infrastructure.

## 2.1 The Science

The scientific foundation of Project Salsabil is based on the relationship between plant water status and land surface temperature. Under normal conditions, healthy crops regulate their temperature through transpiration: water is drawn from the soil, transported through the plant, and released as vapor through the stomata. This evaporative process cools the canopy. When water becomes limited, stomata begin to close, transpiration decreases, and canopy temperature rises. This thermal response provides a measurable signal of crop stress before visual symptoms, such as wilting or leaf discoloration, become apparent.

Satellite remote sensing makes it possible to observe this process over large agricultural areas. Thermal infrared imagery provides estimates of surface temperature, while multispectral data can be used to derive vegetation indices such as NDVI and EVI, which describe plant vigor, biomass, and canopy density. Together with local weather information, including air temperature, humidity, and wind speed, these variables can be combined in surface energy balance models to estimate evapotranspiration and identify water stress. In the Tunisian context, this approach is especially valuable because it allows continuous monitoring of heterogeneous agricultural zones without requiring constant field inspection.

A key advantage of this method is that it is scalable across crop types and regions, provided that the system is properly calibrated using local ground-truth measurements. This is essential for Tunisia, where olives, cereals, vegetables, and orchards each respond differently to heat and water availability. The scientific challenge is therefore not only to detect heat, but to interpret it correctly in relation to the crop, the soil, and the local climate.

## 2.2 The Metric

The primary operational metric is the **Crop Water Stress Index (CWSI)**, which quantifies the degree of water stress experienced by a crop. It is one of the most widely used indicators for thermal-based irrigation monitoring because it provides a normalized measure of plant stress that can be compared across fields and time periods.

The CWSI is defined as follows:

$$\mathrm{CWSI} = \frac{(T_c - T_a) - (T_c - T_a)_{\mathrm{ll}}}{(T_c - T_a)_{\mathrm{ul}} - (T_c - T_a)_{\mathrm{ll}}} \tag{2.1}$$

where:

- $T_c$ is the canopy (leaf surface) temperature [°C],
- $T_a$ is the ambient air temperature [°C],
- $(T_c - T_a)_{\mathrm{ll}}$ is the lower baseline corresponding to a well-watered reference condition,
- $(T_c - T_a)_{\mathrm{ul}}$ is the upper baseline corresponding to a non-transpiring or severely stressed reference condition.

The CWSI value is typically interpreted on a scale from 0 to 1. A value near 0 indicates that the crop is close to a well-watered state, while a value near 1 indicates high water stress. In practice, the index helps answer a very operational question: which fields in Tunisia need water now, and which fields can wait? This makes it especially suitable for precision irrigation management, since it converts raw thermal measurements into a decision-ready signal.

For Project Salsabil, the CWSI is not used as a generic theoretical indicator. It is adapted to Tunisian conditions through crop-specific calibration, local weather integration, and comparison against ground-truth soil moisture data. This is important because the same thermal reading can mean different things depending on the crop, soil type, and region. For example, a thermal rise in a irrigated tomato field in Cap Bon does not have the same agronomic meaning as the same reading in an olive orchard in Sfax. The calibration process therefore ensures that the CWSI reflects real field conditions rather than abstract remote-sensing values.

The metric will also be validated against in situ measurements from sensor stations and field observations. This validation step is essential to establish thresholds for alert generation, to reduce false positives, and to ensure that the system remains reliable under Tunisian climatic variability. Over time, these calibrated thresholds will form the basis of a national thermal stress reference library for major Tunisian crops.

## 2.3 The Delivery

The delivery model of Project Salsabil is organized into three connected layers. The first layer is the national agricultural mapping layer, which builds a spatial database of Tunisian farmland, crop types, and irrigation zones. This layer provides the geographic foundation for identifying where crops are located and what type of thermal behavior is expected in each area.

The second layer is the AI analytics platform. This layer processes satellite imagery, weather data, and ground-truth inputs to calculate water stress indicators and detect thermal anomalies. Its role is to transform raw data into actionable intelligence, such as stress maps, irrigation recommendations, and early warnings for regional authorities. At this stage, the platform acts as the intelligence core of the system, supporting both farmers and decision-makers with field-level and regional-level insights.

The third layer is the long-term Low Earth Orbit satellite constellation. This layer represents Tunisia's sovereign data future. Instead of relying permanently on external providers, the country would own its observation infrastructure and control the acquisition schedule, revisit frequency, and data pipeline. This enables faster monitoring, better territorial coverage, and long-term strategic independence. In the final architecture, the constellation is not only a technical upgrade; it is the mechanism that ensures that Tunisia owns the full chain of agricultural intelligence from observation to decision.

Together, these three layers create a complete delivery pipeline: spatial mapping, analytical interpretation, and sovereign data generation. This structure allows Project Salsabil to evolve gradually, starting with rented satellite data and ground calibration, then progressing toward a fully national agricultural intelligence system.

# Chapter 3

# Phase 1: National Agricultural Mapping & Calibration

Phase 1 establishes the foundational geospatial and agronomic understanding of Tunisia's agricultural landscape through the construction of a national agricultural digital twin. The objective is to build a continuously updated, data-driven representation of crop conditions, environmental drivers, and soil-water dynamics across the country.

This digital twin integrates multi-source Earth observation data with agronomic and climatic datasets to simulate the real-time state of agricultural water stress across Tunisia. It serves as the calibration backbone for subsequent machine learning and CubeSat-based monitoring systems described in later phases.

This phase is entirely ground- and satellite-based, relying on existing Earth observation systems rather than new hardware deployments. Its output is a harmonized, multi-layer digital twin aligned with Tunisia's administrative and ecological structure (governorates, soil zones, and crop regions), as summarized in Table 6.1.

## 3.1 Mapping

The agricultural digital twin is constructed using multi-source satellite imagery, primarily:

- **Sentinel-2 (10–20 m resolution, 5-day revisit)** for high-resolution vegetation monitoring
- **Landsat-9 (30 m resolution, 16-day revisit)** for long-term historical consistency
- **MODIS (250–1000 m resolution, daily revisit)** for high-frequency temporal dynamics

These datasets are fused to construct a continuous spatio-temporal representation of crop health across Tunisia. Vegetation indices such as NDVI, SAVI, and EVI are computed to quantify canopy vigor, while land surface temperature (LST) is derived from thermal infrared bands to estimate canopy heat stress.

The resulting system is a dynamic agricultural digital twin, where each pixel is associated with a time-evolving stress state rather than a static observation., where each pixel is associated with a temporal stress signature rather than a single snapshot value. This enables early detection of drought patterns and localized water stress anomalies.

Spatial aggregation is performed at the level of Tunisian governorates to align with administrative decision-making structures, enabling policy-relevant visualization and intervention planning.

## 3.2 Baseline Data

To support robust model calibration, a comprehensive set of baseline environmental and agronomic datasets is compiled. These include:

- **Historical evapotranspiration ($ET_0$)**: derived from FAO Penman–Monteith reanalysis datasets
- **Soil classification maps**: defining soil texture classes (sandy, loam, clay-loam) and associated hydrological properties
- **Crop calendars**: seasonal planting and harvesting schedules for major Tunisian crops (wheat, barley, olive, tomato, citrus, potato, pepper)
- **Irrigation infrastructure layers**: mapping irrigated vs rainfed agricultural zones
- **Climatic normals**: long-term averages of temperature, humidity, solar radiation, and wind speed

These datasets are harmonized into a unified geospatial database, enabling feature alignment across spatial, temporal, and agronomic dimensions. This baseline layer provides the contextual priors required for machine learning models to distinguish between natural climatic variation and abnormal crop stress signals.

## 3.3 Ground-Truth

Ground-truth data collection is designed to validate satellite-derived stress indicators and calibrate the Crop Water Stress Index (CWSI) model.

The validation framework includes three complementary components:

### In-situ Sensor Networks

Distributed soil moisture and temperature sensors are deployed across representative agricultural zones in Tunisia, covering diverse climatic gradients from humid northern regions (e.g., Béja) to arid southern regions (e.g., Gafsa). These sensors provide continuous measurements of:

- Volumetric soil moisture ($m^3/m^3$)
- Air temperature at canopy level
- Relative humidity and microclimatic variation

### Reference Measurement Sites

A subset of fields is equipped with higher-precision validation instruments, including eddy-covariance systems where feasible, to estimate evapotranspiration fluxes and validate energy balance assumptions used in satellite interpretation.

### Farmer Field Surveys

Structured surveys are conducted with local farmers to record irrigation events, crop type verification, and perceived stress conditions. These qualitative observations are used to validate and correct satellite-based stress classifications.

All ground-truth data is synchronized with satellite acquisition timestamps (2021–2023 dataset window), ensuring temporal alignment between observed and predicted stress conditions.

The final output of Phase 1 is a validated, multi-layer agricultural dataset for Tunisia, serving as the calibration backbone for the machine learning models used in Phase 2.

# Chapter 4

# Phase 2: AI Development & Commercial Pilot

Phase 2 transforms the calibrated geospatial foundation from Phase 1 into an operational artificial intelligence system capable of estimating crop water stress in near real-time. This phase bridges satellite observation, environmental modeling, and decision support into a unified predictive engine tailored for Tunisia's agricultural conditions.

The primary objective is to develop a scalable Crop Water Stress Index (CWSI) prediction system and deploy it as a pilot commercial platform for farmers, agricultural agencies, and water resource managers.

## 4.1 The Engine

The core of the system is a physics-informed machine learning model designed to estimate the Crop Water Stress Index (CWSI) from multi-source environmental inputs.

The model integrates three complementary components:

- **Tree-based ensemble models (XGBoost / Random Forest)** for structured tabular data (soil, weather, crop type, region)
- **Temporal encoding layers** to capture seasonal dynamics (month, year, phenological cycles)
- **Feature weighting guided by agronomic physics**, ensuring interpretability of stress drivers

Given the tabular and multi-modal nature of the dataset, gradient-boosted decision trees are prioritized due to their robustness, interpretability, and strong performance on heterogeneous agricultural datasets.

The target variable is the continuous Crop Water Stress Index (CWSI), ranging from 0 (no stress) to 1 (critical stress). A secondary classification head produces categorical stress levels (Healthy, Mild, Moderate, Severe) for operational decision-making.

The training pipeline includes:

- Data normalization per environmental zone
- Handling of missing satellite observations via temporal interpolation
- Feature importance regularization to prevent overfitting on LST-dominated signals
- Cross-validation stratified by governorate to ensure geographic generalization across Tunisia

Inference is optimized for cloud deployment, enabling batch processing of satellite-derived feature grids and real-time API-based predictions for individual farms.

## 4.2 Data Fusion

The system relies on multi-source data fusion to construct a unified representation of crop water stress drivers.

The fused data streams include:

- **Optical satellite data**: NDVI, SAVI, EVI (vegetation health indicators)
- **Thermal satellite data**: Land Surface Temperature (LST) as a proxy for canopy stress
- **Weather reanalysis data**: air temperature, humidity, wind speed, solar radiation, VPD, $ET_0$
- **Soil properties**: soil type, field capacity, wilting point, soil moisture estimates
- **IoT / ground sensors (optional extension)**: in-situ validation of soil moisture and microclimate conditions

All inputs are synchronized temporally and spatially at the field level using geospatial indexing. Each observation is mapped to a unified feature vector representing the physical state of a farm at a given time.

A key aspect of the fusion process is the separation between:

- **slow-changing variables** (soil type, crop type, region)
- **fast-changing variables** (weather, LST, vegetation indices)

This separation improves model stability and prevents noise amplification from short-term atmospheric fluctuations.

The final fused dataset corresponds directly to the structured feature schema defined, enabling seamless training and deployment consistency.

## 4.3 Interface

The system is exposed through a dual-layer interface designed for both farmers and agricultural decision-makers.

### National Dashboard

A geospatial dashboard provides a Tunisia-wide overview of crop water stress. Governorates are color-coded on a gradient from blue (low stress) to orange/red (high stress), representing the aggregated mean CWSI of all farms within each region.

Users can drill down from governorate-level views into individual farms represented as spatial nodes. Each farm node displays:

- Owner information
- Crop type and cultivated area
- Current stress level (CWSI and categorical label)
- Historical stress trends

A national statistics panel summarizes:

- Average national stress level
- Regions with highest stress requiring intervention
- Regions with low stress potentially suitable for water redistribution strategies

### Alerting System

When localized stress conditions exceed predefined thresholds, the system triggers automated recommendations. In non-critical scenarios, farmers receive advisory notifications encouraging irrigation optimization.

Communication channels include:

- SMS alerts for rural accessibility
- Web dashboard notifications
- API endpoints for integration with agricultural agencies

### Prototype Implementation

For the purposes of this prototype, the geospatial map is implemented as an interactive grid-based abstraction of Tunisia, where each region is represented as a clickable tile rather than a high-resolution geographic map. This simplifies visualization while preserving the spatial logic of the system.

The frontend is implemented using a React-based stack with a Tailwind CSS design system, enabling responsive visualization of stress distributions and farm-level interactions.

# Chapter 5

# Phase 3: Sovereign LEO Constellation Launch

Phase 3 of Project Salsabil represents the transition from a data-dependent system to a fully sovereign agricultural intelligence infrastructure. At this stage, Tunisia moves from consuming external satellite products to operating its own dedicated Low Earth Orbit (LEO) constellation optimized specifically for agricultural monitoring, thermal stress detection, and irrigation intelligence. This layer ensures full control over revisit frequency, spatial resolution, and data pipelines, enabling real-time decision-making at a national scale.

## 5.1 The Hardware

The satellite platform is based on a standardized 6U CubeSat architecture, selected for its balance between cost, scalability, and payload capacity. Each unit is designed to operate in the 10–12 kg mass class, with an average power budget of approximately 20–40 W depending on orbital phase and payload operation mode.

The Attitude Determination and Control System (ADCS) is based on a combination of reaction wheels, magnetorquers, and star-tracking for fine pointing accuracy. This is essential to ensure stable imaging for thermal infrared acquisition over small agricultural parcels typical of Tunisian farmland.

Communication is handled through an S-band or X-band downlink, depending on mission phase and ground station availability. The system is designed to be compatible with standard rideshare launch opportunities, including Falcon 9 Transporter missions or equivalent European launch providers such as Vega-C, ensuring deployment flexibility and cost optimization.

The satellite bus is designed with modularity in mind, allowing future upgrades to payload sensors or onboard processing units without redesigning the entire platform.

## 5.2 Payload

The primary payload consists of a Long-Wave Infrared (LWIR) thermal imaging system, designed for land surface temperature acquisition and crop water stress monitoring. This sensor is the key enabler for detecting evapotranspiration-related temperature variations in agricultural zones across Tunisia.

In addition to the thermal payload, a multispectral camera is included to compute vegetation indices such as NDVI and NDWI. These indices provide complementary information on plant vigor, chlorophyll content, and biomass density, improving the robustness of downstream water stress estimation models.

Data acquired by the payload is transmitted to ground infrastructure via a LoRaWAN-based communication architecture, aligned with existing IoT deployments in Tunisia (notably initiatives previously implemented by operators such as Telnet). This approach allows integration between satellite observations and terrestrial sensor networks within a unified national data ecosystem.

The target spatial resolution is in the range of 10 to 30 meters per pixel, depending on orbital altitude and imaging configuration. Swath width is optimized to ensure sufficient coverage of Tunisian agricultural regions within a high revisit cycle.

Radiometric calibration remains a critical requirement to ensure consistency across temporal observations, enabling reliable comparison of crop stress indicators across seasons, regions, and crop types.

## 5.3 Onboard AI

In this architecture, the CubeSats do not perform onboard artificial intelligence processing. Instead, their role is strictly limited to data acquisition, preprocessing, and transmission.

Each satellite captures raw thermal and multispectral imagery and performs minimal onboard operations such as sensor calibration, basic compression, and packet formatting to optimize transmission efficiency.

The full analytical pipeline, including Crop Water Stress Index (CWSI) computation, anomaly detection, and irrigation recommendation generation, is executed entirely on-ground within national processing infrastructure.

This design choice is intentional and aligns with three key constraints:

- **Computational constraint:** CubeSats have limited power and processing capacity.
- **Maintainability:** On-ground updates to AI models are easier and more flexible.

- **Sovereignty:** All intelligence remains under national control within Tunisian infrastructure.

The ground segment aggregates satellite data with weather stations and IoT soil moisture networks to perform full surface energy balance modeling and generate operational irrigation insights.

## 5.4 The Orbit

The constellation operates in a Sun-Synchronous Orbit (SSO) at an altitude of approximately 500–600 km. This orbit is selected to ensure consistent lighting conditions across imaging passes, which is essential for reliable thermal and multispectral comparisons over time.

The Local Time of Ascending Node (LTAN) is chosen to optimize agricultural observation windows, typically aligned with mid-morning conditions where thermal contrast between stressed and healthy vegetation is most detectable.

The constellation design targets a revisit time of approximately 4 hours over Tunisian territory under full deployment. This requires a multi-satellite configuration distributed across orbital planes to ensure temporal continuity.

The number of satellites in the initial operational constellation is estimated between 4 and 8 units, scalable depending on coverage requirements and budget constraints.

## 5.5 The Exit

At the end of their operational life, all satellites are designed to comply with international orbital debris mitigation standards, including guidelines from the Inter-Agency Space Debris Coordination Committee (IADC).

Each satellite includes a passive or controlled deorbit mechanism to ensure atmospheric re-entry within a maximum of 5 years after end-of-mission. This may include atmospheric drag augmentation devices or controlled propulsion burns, depending on system configuration.

The deorbit strategy is designed to minimize long-term orbital debris risk while ensuring full compliance with international sustainability standards. This reflects the project's commitment not only to technological sovereignty but also to responsible space operations.

Ultimately, Phase 3 is not only an infrastructure milestone but a strategic transition: it ensures that Tunisia maintains long-term autonomy over agricultural intelligence while contributing responsibly to the shared orbital environment.

# Chapter 6

# Implementation Roadmap & Financial Estimates

This chapter outlines the phased implementation strategy of Project Salsabil, spanning from national-scale calibration to full satellite-enabled operational deployment. The roadmap is designed to progressively reduce technical uncertainty while maximizing early agricultural impact in Tunisia.

Each phase builds upon the previous one, transitioning from ground-based observation to AI-driven decision systems and ultimately to a dedicated space-based sensing infrastructure.

**Table 6.1:** Project Salsabil – Implementation Roadmap & Cost Estimates

| Phase | Timeline | Key Activities | Estimated Cost |
|---|---|---|---|
| **Phase 1 Mapping & Calibration** | Q1–Q4 Y1 | National-scale agricultural mapping using Sentinel-2, Landsat-9, and MODIS data; construction of baseline environmental datasets; soil and crop classification harmonization; ground-truth validation campaign across Tunisian governorates; calibration of initial Crop Water Stress Index (CWSI) models. | \$1.2M – \$2.0M |
| **Phase 2 AI & Commercial Pilot** | Q1–Q4 Y2 | Development of AI-based CWSI prediction engine; training of physics-informed machine learning models; cloud deployment of inference pipeline; integration of multi-source data streams (satellite, weather, soil); development of farmer-facing dashboard and alerting system (SMS, web, API); pilot deployment in selected agricultural regions of Tunisia. | \$2.5M – \$4.0M |
| **Phase 3 LEO Constellation** | Y3–Y5 | Design and development of CubeSat-based remote sensing system; integration of onboard multispectral and thermal payloads; LoRaWAN ground station network deployment; launch campaign and orbital commissioning; establishment of operational satellite constellation for high-frequency agricultural monitoring. | \$20M – \$45M |
| **Total** | 5 years | Full transition from terrestrial calibration system to autonomous space-based agricultural intelligence infrastructure. | \$23.7M – \$51M |

*Note: Cost estimates are based on comparable Earth observation systems, CubeSat development benchmarks, and cloud AI infrastructure scaling assumptions.*

## 6.1 Long-Term Maintainability

The long-term sustainability of Project Salsabil is ensured through a hybrid operational model combining satellite infrastructure, cloud-based analytics, and national agricultural integration.

#### Satellite and Ground Segment Maintenance

The CubeSat constellation is designed with a replenishment cycle of 3–5 years, consistent with standard low-Earth orbit small satellite lifetimes. Ground stations using LoRaWAN-enabled reception nodes are deployed across Tunisia and maintained through partnerships with local telecom and research institutions.

#### AI Model Lifecycle Management

The CWSI prediction models are continuously retrained using newly acquired satellite and ground-truth data. A quarterly retraining cycle is implemented to account for:

- Climate variability and drought anomalies
- Crop rotation and seasonal changes
- Sensor drift and calibration updates

#### Institutional Partnerships

Operational sustainability is reinforced through collaboration with national and regional stakeholders, including agricultural ministries, universities, and space agencies. These partnerships ensure data sovereignty, infrastructure access, and long-term operational continuity.

#### Data Governance

All agricultural data generated within the system remains under national ownership, with controlled access layers for researchers, policymakers, and farmers. Open-data subsets may be released for scientific research under regulated conditions.

## 6.2 Economic Return on Investment

The economic viability of Project Salsabil is evaluated through direct agricultural savings, yield improvements, and national-scale water optimization benefits.

### Water Savings

By enabling precision irrigation scheduling based on real-time CWSI estimates, the system is projected to reduce agricultural water consumption by 15–30% in irrigated regions. This corresponds to millions of cubic meters of water saved annually in Tunisia, a country facing increasing water scarcity.

### Yield Improvement

Early detection of crop stress conditions allows farmers to intervene before irreversible damage occurs. Conservative estimates indicate a 5–12% increase in yield stability for high-value crops such as tomatoes, citrus, and vegetables.

### National Water Efficiency

At the policy level, aggregated stress maps enable optimized redistribution of water resources across governorates. Regions identified as low-stress zones can support adaptive irrigation strategies, reducing overall national water stress during peak drought periods.

### Cost Recovery Model

The system operates under a hybrid public-service and data-platform model:

- Subscription-based access for agribusinesses and cooperatives
- Government licensing for national water management systems
- API-based data services for research and agricultural analytics companies

### Sensitivity Analysis

Economic returns are highly sensitive to drought severity and adoption rate. In high-adoption scenarios (>60% of irrigated land), the system achieves full cost recovery within 4–6 years through combined water savings and yield gains.

Overall, Project Salsabil demonstrates strong long-term return on investment potential while simultaneously addressing critical water security challenges in Tunisia.

# Chapter 7

# Prototype Implementation

This chapter presents the implementation of the Project Salsabil prototype, detailing the system architecture, individual components, and overall workflow. The prototype demonstrates the feasibility of integrating satellite-inspired data, machine learning, and decision-support interfaces for agricultural water stress monitoring in Tunisia.

## 7.1 System Architecture

The prototype follows a modular architecture composed of five main components: the Digital Twin, AI Model, Frontend Dashboard, Client-Server backend, and SMS alert system.

#### Technology Stack

| **Component** | **Technology** |
|---|---|
| Digital Twin | Python, Streamlit |
| AI Model | Python, Scikit-learn |
| Frontend Dashboard | React, Vite, Tailwind CSS |
| Client-Server | Python (Flask) |
| SMS Alerts | Twilio API (planned) |

**Table 7.1:** Prototype Technology Stack

The system is structured into three layers:

- **Data Layer**: simulated and partially real agricultural datasets
- **Processing Layer**: Digital Twin and AI model
- **Application Layer**: dashboard and alert system

**Architecture Diagram:** A system diagram illustrating data flow from acquisition to decision-making is included as a figure.

## 7.2 Frontend Dashboard

The frontend dashboard provides an interactive visualization of agricultural stress across Tunisia.

#### Features

- Grid-based representation of Tunisia (prototype abstraction)
- Governorates represented as colored blocks:
  - Blue: low stress
  - Orange: high stress
- Clickable regions showing farms as nodes
- Farm-level modal displaying:
  - Owner
  - Crop type
  - Area
  - Stress level (CWSI)
- National statistics panel:
  - Average stress level
  - High-stress regions
  - Low-stress regions (water reallocation candidates)

#### Technologies

The dashboard is implemented using React for UI logic, Vite for development tooling, and Tailwind CSS for styling.


**Figure 7.1:** Frontend Dashboard.

## 7.3 Digital Twin

The Digital Twin represents a simplified, dynamic model of Tunisia's agricultural system.

### Functionality

- Simulation of agricultural stress at regional and farm levels
- Aggregation of environmental variables into structured datasets
- Generation of grid-based stress maps

### Modeled Variables

- Soil moisture
- Vegetation indices (NDVI, SAVI, EVI)
- Land Surface Temperature (LST)
- Weather parameters

### Technologies

- Python
- Streamlit
- Pandas and NumPy

Future improvements include integration of GeoPandas and Rasterio for real geospatial processing.

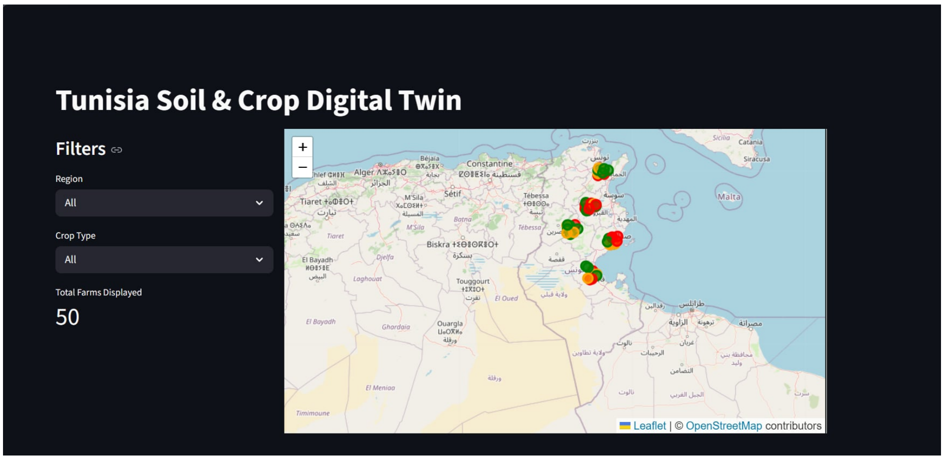


**Figure 7.2:** Digital Twin.

## 7.4 AI Model

The AI model estimates crop water stress using environmental and satellite-derived features.

### Model Type

Tree-based models such as Random Forest and Gradient Boosting are used due to their robustness with tabular agricultural data.

### Inputs

- Land Surface Temperature (LST)
- NDVI, SAVI, EVI
- Air temperature
- Relative humidity
- Wind speed
- Solar radiation
- Vapor Pressure Deficit (VPD)
- Soil moisture
- Soil type

- Crop type
- Temporal features (month, year)

### Outputs

- Continuous CWSI score (0–1)
- Stress classification:
  - Healthy
  - Mild stress
  - Moderate stress
  - Severe stress

```python
# Model hyperparameters (optimized for regression)
MODEL_PARAMS = {
    'n_estimators': 200,            # Number of trees
    'max_depth': None,              # No depth limit for full learning
    'min_samples_split': 2,         # Minimum samples to split
    'min_samples_leaf': 1,          # Minimum samples per leaf
    'max_features': 'sqrt',         # Feature subset size
    'random_state': RANDOM_STATE,   # Reproducibility
    'n_jobs': -1,                   # Use all CPU cores
    'bootstrap': True               # Bootstrap sampling
}

print("  Initializing Random Forest Regressor with parameters:")
for param, value in MODEL_PARAMS.items():
    print(f"   • {param}: {value}")

# Initialize and train the model
model = RandomForestRegressor(**MODEL_PARAMS)
print(f"\n Training model on {X_train.shape[0]} samples...")
model.fit(X_train, y_train)
print(" Model training completed")
```
0.5s Python

**Figure 7.3:** AI Model.

## 7.5 Client-Server & SMS Alerts

### Backend

The backend is implemented using Flask, providing API endpoints for communication between the frontend dashboard and the AI model.

### SMS Alert System

The SMS alert system is based on the Twilio API (planned integration).

### Alert Logic

- CWSI > 0.5: Moderate stress alert
- CWSI > 0.75: Critical stress alert

#### Example Message

> Alert: Your field (Tomato, Kairouan) is experiencing high water stress (CWSI = 0.78). Immediate irrigation is recommended.

## 7.6 End-to-End Workflow

The prototype operates through the following pipeline:

1. Environmental and satellite-like data are generated or collected
2. Data is processed within the Digital Twin layer
3. The AI model predicts CWSI values
4. Results are transmitted to the backend server
5. The dashboard visualizes stress levels
6. SMS alerts are triggered when thresholds are exceeded

## 7.7 Prototype Summary

| **Component** | **Status** | **Notes** |
|---|---|---|
| Frontend Dashboard | Work in Progress | Grid-based Tunisia map, not fully styled or connecte |
| Digital Twin | Work in Progress | Functional but limited by data availability |
| AI Model | Work in Progress | Model defined, limited training data |
| Client-Server | Complete (Untested) | API implemented but not fully integrated |
| SMS Alerts | Work in Progress | Twilio integration planned, not tested |

**Table 7.2:** Prototype Development Status

Several figures will be included to illustrate the prototype components, including dashboard screenshots, digital twin visualizations, AI outputs, and system architecture diagrams.

# Chapter 8

# Conclusion

Project Salsabil provides a strategic roadmap for addressing Tunisia's water deficit through technological sovereignty and precision agriculture. By shifting from reactive, uniform irrigation to a proactive, data-driven model, the system enables the detection of "thermal anomalies" before crop damage becomes irreversible.

The three-phase implementation strategy—spanning from national mapping to the launch of a sovereign LEO constellation—ensures that Tunisia eliminates dependence on foreign data providers while gaining full control over its agricultural intelligence. With a projected 15–30% reduction in water consumption and a cost recovery period of 4–6 years in high-adoption scenarios, the project demonstrates both environmental necessity and economic viability. Ultimately, Project Salsabil establishes the foundation for a climate-resilient agricultural sector, securing food stability and strategic autonomy for Tunisia's future.

# Appendix A

# Acronyms & Abbreviations

| **Acronym** | **Definition** |
|---|---|
| ADCS | Attitude Determination and Control System |
| AI | Artificial Intelligence |
| AIT | Assembly, Integration, and Testing |
| API | Application Programming Interface |
| CRDA | Regional Commissariat for Agricultural Development (Commissariat Régional au Déve |
| CWSI | Crop Water Stress Index |
| $ET_0$ | Reference Evapotranspiration |
| EVI | Enhanced Vegetation Index |
| GNSS-RO | GNSS Radio Occultation |
| GSD | Ground Sampling Distance |
| IADC | Inter-Agency Space Debris Coordination Committee |
| IoT | Internet of Things |
| KPI | Key Performance Indicator |
| LEO | Low Earth Orbit |
| LST | Land Surface Temperature |
| LTAN | Local Time of Ascending Node |
| LWIR | Long-Wave Infrared |
| ML | Machine Learning |
| MODIS | Moderate Resolution Imaging Spectroradiometer |
| NDVI | Normalised Difference Vegetation Index |
| SaaS | Software as a Service |
| SAR | Synthetic Aperture Radar |
| SAVI | Soil Adjusted Vegetation Index |
| SSO | Sun-Synchronous Orbit |
| TIR | Thermal Infrared |
| VPD | Vapor Pressure Deficit |